IAM Ticket IAM-0004: MFA Support: Lost Phone

AccessGuard Health – Healthcare IAM Operations Simulation

Ticket ID	IAM-0004
Related	Board task Activity 4 — MFA Support; Step 10 of the Entra ID hands-on (Invoke-HvmcMfaSupport.ps1)
Request type	MFA support: lost device (Microsoft Authenticator)
Priority	High (the user cannot sign in, and the lost phone could be used to approve sign-ins)
Status	Resolved and closed
Date	November 5, 2026
Opened / closed	November 5, 2026, 3:25 PM / November 5, 2026, 3:54 PM
Requester	Dr. Sarah Chen, Physician, Doctors (fictional)
Approver	IT Security
Handled by	Sabitha S Karanavar (IT), using the help-desk account helpdesk.admin (Authentication Administrator + Reports Reader)
Employee	Sarah Chen (HVMC-1001), Doctors (fictional)

Request

Dr. Chen called the help desk: she lost her phone this morning and cannot approve Microsoft Authenticator sign-in requests. She needs access to the clinical apps for her shift and has a new phone ready.

Identity Verification

The caller's identity was confirmed before any change, because help-desk calls are a known way for attackers to take over accounts (for example, the 2023 MGM Resorts attack began with a call to the help desk).

Method	Call-back to the phone number HR has on file
Result	Identity confirmed before the reset

Systems and Account Changes

Item	Before	After
System	Microsoft Entra ID (HVMC tenant), security defaults on	Same

Item	Before	After
Username	sarah.chen	sarah.chen
Account state	Enabled	Enabled
MFA methods	Microsoft Authenticator (device: iPhone 12 Pro) (old phone)	Removed at reset; then Microsoft Authenticator (device: iPhone 12 Pro) newly registered by the user (new phone)
Sign-in sessions	Active, including on the lost phone	Revoked at 20:27 UTC
Password	Set	Reset by the IAM admin during the lab only (the test password was not at hand), then changed by the user. Not part of the MFA reset; in a real case the password is left alone.
Group and app access	SG-Doctors-EHR-RW and its apps	Unchanged

Actions Taken

- Ticket opened; recorded the before state of the account and its MFA methods (read-only Status run).
- Verified the caller's identity before any change: Call-back to the phone number HR has on file.
- Ran the reset as a dry run first and checked the plan.
- 20:27:49 UTC — revoke all sign-in sessions: ok
- 20:27:52 UTC — remove Microsoft Authenticator device: iPhone 12 Pro: ok
- Asked Dr. Chen to sign in and register her new phone on her Security info page; she did it herself.
- Ran the Verify check and reviewed the audit log, then closed the ticket.

Verification

Check	Expected	Result
identity verified before the reset	Yes	Yes
old MFA methods removed at reset	Yes	Yes
new MFA method registered after the reset	Yes	Yes
audit: admin deleted security info	Yes	Yes
audit: user registered security info	Yes	Yes
account still enabled	Yes	Yes

Audit trail (Entra ID)

Time (UTC)	Activity	Result	By
2026-11-05 20:27	Update StsRefreshTokenValidFrom Timestamp	success	helpdesk.admin@<tenant>
2026-11-05 20:27	Update user	success	helpdesk.admin@<tenant>
2026-11-05 20:27	Update user	success	Azure Credential Configuration Endpoint Service
2026-11-05 20:27	Update user	success	Azure Credential Configuration Endpoint Service
2026-11-05 20:27	Admin deleted security info	success	helpdesk.admin@<tenant>
2026-11-05 20:49	Update PasswordProfile	success	Microsoft password reset service
2026-11-05 20:49	Update user	success	Microsoft password reset service
2026-11-05 20:49	Update PasswordProfile	success	Microsoft password reset service
2026-11-05 20:49	Reset user password	success	Microsoft password reset service
2026-11-05 20:49	Update StsRefreshTokenValidFrom Timestamp	success	Microsoft password reset service
2026-11-05 20:49	Reset password (by admin)	success	sabithaskaranavar@<tenant>
2026-11-05 20:51	Update PasswordProfile	success	Microsoft password reset service
2026-11-05 20:51	Update user	success	Microsoft password reset service
2026-11-05 20:51	Update PasswordProfile	success	Microsoft password reset service
2026-11-05 20:51	Reset user password	success	Microsoft password reset service
2026-11-05 20:51	Update StsRefreshTokenValidFrom Timestamp	success	Microsoft password reset service
2026-11-05 20:51	Reset password (by admin)	success	sabithaskaranavar@<tenant>
2026-11-05 20:51	Update PasswordProfile	success	Microsoft password reset service

Time (UTC)	Activity	Result	By
2026-11-05 20:51	Change user password	success	Microsoft password reset service
2026-11-05 20:51	Update StsRefreshTokenValidFrom Timestamp	success	Microsoft password reset service
2026-11-05 20:51	Change password (self-service)	success	sarah.chen@<tenant>
2026-11-05 20:52	Update user	success	Azure MFA StrongAuthenticationService
2026-11-05 20:52	Update user	success	Azure MFA StrongAuthenticationService
2026-11-05 20:52	Update user	success	Azure Credential Configuration Endpoint Service
2026-11-05 20:52	User registered security info	success	sarah.chen@<tenant>
2026-11-05 20:52	Update user	success	Azure Credential Configuration Endpoint Service

Resolution Notes

- Sessions were revoked first, so anyone holding the lost phone was signed out before the MFA method was removed.
- Only the MFA methods were removed; the password and the password-reset email were kept, so the user could sign in and register again straight away.
- The user registered the new phone herself; the help desk never handled her credentials.
- The Entra ID audit log records both the admin removal and the user's new registration, which is the evidence for this ticket.

Lessons learned

- A Temporary Access Pass (a time-limited, one-time code issued by the help desk after the identity check) would let the user register the new phone without relying on the password alone. It is worth evaluating for HVMC's MFA-recovery process.
- The identity check is the most important control in this process: a reset without it would hand the account to an attacker.
- The reset was done by a dedicated help-desk account with only the Authentication Administrator and Reports Reader roles, not by the Global Admin. That role can reset MFA for normal users but never for admins (least privilege, separation of duties).
- Security defaults blocked the personal-account Global Admin from Microsoft Graph PowerShell (AADSTS530035). Security defaults were kept on; the dedicated in-tenant help-desk account was the fix.

- Entra ID can still list a deleted method for a short time. The script's first re-check 3 seconds after the reset was too early; it now waits up to 2 minutes, and a Status check two minutes later confirmed MFA methods: 0.
- Lab note: Sarah's password was also reset by the IAM admin during the lab because the test password was not at hand. In a real lost-phone case the password is not touched.

Evidence

Evidence files are kept locally in step10_mfa_support/evidence (not on GitHub). Phone numbers and directory IDs are not shown.

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Status
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-10-29 19:27 UTC (23):
    2026-10-29 20:39  Update PasswordProfile                 success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                            success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                 success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                    success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update StsRefreshTokenValidFrom Timestamp success  by Global Admin (IAM admin)
    2026-11-02 19:13  Update user                            success  by Global Admin (IAM admin)
    2026-11-04 17:25  Update PasswordProfile                 success  by Microsoft password reset service
    2026-11-04 17:25  Update user                            success  by Microsoft password reset service
    2026-11-04 17:25  Update PasswordProfile                 success  by Microsoft password reset service
    2026-11-04 17:25  Reset user password                    success  by Microsoft password reset service
    2026-11-04 17:25  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:25  Reset password (by admin)              success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)         failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                 success  by Microsoft password reset service
    2026-11-04 17:27  Change user password                   success  by Microsoft password reset service
    2026-11-04 17:27  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:27  Change password (self-service)         success  by sarah.chen@<tenant>
    2026-11-04 17:30  Update user                            success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:30  Update user                            success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                            success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                            success  by Azure MFA StrongAuthenticationService
    2026-11-04 17:34  Update user                            success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info          success  by sarah.chen@<tenant>
```

Figure 1: Before: Sarah's registered methods (Microsoft Authenticator on the old phone)

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Reset
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-10-29 19:27 UTC (23):
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                             success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                     success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update StsRefreshTokenValidFrom Timestamp success  by Global Admin (IAM admin)
    2026-11-02 19:13  Update user                             success  by Global Admin (IAM admin)
    2026-11-04 17:25  Update PasswordProfile                  success  by Microsoft password reset service
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    2026-11-04 17:25  Reset password (by admin)               success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)          failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                  success  by Microsoft password reset service
    2026-11-04 17:27  Change user password                    success  by Microsoft password reset service
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    2026-11-04 17:30  Update user                             success  by Azure MFA StrongAuthenticationService
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    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
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    2026-11-04 17:34  Update user                             success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info           success  by sarah.chen@<tenant>

DRY RUN  Lost-device reset for Sarah Chen
   1. revoke all sign-in sessions (refresh tokens on every device, including the lost phone)
   2. remove Microsoft Authenticator device: iPhone 12 Pro
      (kept: password, and the password-reset email)

DRY RUN - nothing was changed. Verify the caller's identity, then run with -Apply -VerifiedBy "...".
```

Figure 2: Dry run: the planned steps, nothing changed

```
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-10-29 19:27 UTC (23):
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                             success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                     success  by Global Admin (IAM admin)
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    2026-11-04 17:25  Reset user password                     success  by Microsoft password reset service
    2026-11-04 17:25  Update StsRefreshTokenValidFrom Timestamp success  by Microsoft password reset service
    2026-11-04 17:25  Reset password (by admin)               success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)          failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                  success  by Microsoft password reset service
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    2026-11-04 17:30  Update user                             success  by Azure MFA StrongAuthenticationService
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    2026-11-04 17:34  Update user                             success  by Azure MFA StrongAuthenticationService
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    2026-11-04 17:34  Update user                             success  by Azure Credential Configuration Endpoint Service
    2026-11-04 17:34  User registered security info           success  by sarah.chen@<tenant>

APPLY  Lost-device reset for Sarah Chen
   1. revoke all sign-in sessions (refresh tokens on every device, including the lost phone)
   2. remove Microsoft Authenticator device: iPhone 12 Pro
      (kept: password, and the password-reset email)
  20:27:49  revoke all sign-in sessions                          ok
  20:27:52  remove Microsoft Authenticator device: iPhone 12 Pro  ok

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-11-05 20:26 UTC (0):

RESET INCOMPLETE - see the FAILED lines above.
```

Figure 3: Reset applied after the identity check: sessions revoked, Authenticator removed

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Status
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (1):
    Password
  MFA methods: 0  (the user will be asked to register at next sign-in)
  Audit events since 2026-10-29 19:29 UTC (26):
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update user                             success  by Global Admin (IAM admin)
    2026-10-29 20:39  Update PasswordProfile                  success  by Global Admin (IAM admin)
    2026-10-29 20:39  Reset user password                     success  by Global Admin (IAM admin)
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    2026-11-04 17:25  Reset password (by admin)               success  by sabithaskaranavar@<tenant>
    2026-11-04 17:27  Change password (self-service)          failure  by sarah.chen@<tenant>
    2026-11-04 17:27  Update PasswordProfile                  success  by Microsoft password reset service
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    2026-11-04 17:34  User registered security info           success  by sarah.chen@<tenant>
    2026-11-05 20:27  Update StsRefreshTokenValidFrom Timestamp success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                             success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                             success  by Azure Credential Configuration Endpoint Service
```

Figure 4: Status two minutes after the reset: only the password is left, MFA methods: 0, audit shows the session revocation by helpdesk.admin

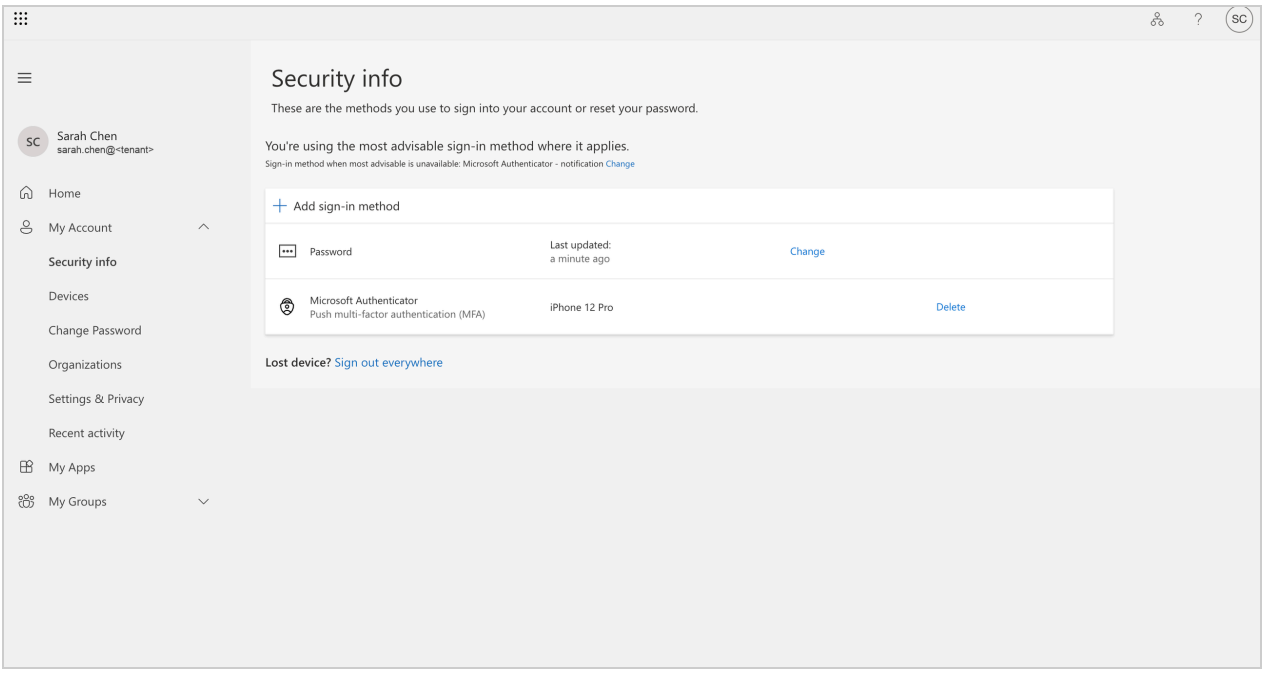


Figure 5: Sarah registers Microsoft Authenticator on her new phone (Security info page)

```
PS /Users/Adult/Downloads/HVMC_IAM_Entra_Handson/step10_mfa_support> ./Invoke-HvmcMfaSupport.ps1 -Action Verify
Connected to Microsoft Graph as helpdesk.admin@<tenant>

IAM-0004  Sarah Chen (HVMC-1001)  sign-in enabled: True
  Registered methods (2):
    Password
    Microsoft Authenticator      device: iPhone 12 Pro
  MFA methods: 1
  Audit events since 2026-11-05 20:26 UTC (26):
    2026-11-05 20:27  Update StsRefreshTokenValidFrom Timestamp success  by helpdesk.admin@<tenant>
    2026-11-05 20:27  Update user                                success  by Azure Credential Configuration Endpoint Service
    2026-11-05 20:27  Update user                                success  by Azure Credential Configuration Endpoint Service
    2026-11-05 20:27  Admin deleted security info                success  by helpdesk.admin@<tenant>
    2026-11-05 20:49  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:49  Update user                                success  by Microsoft password reset service
    2026-11-05 20:49  Update PasswordProfile                     success  by Microsoft password reset service
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    2026-11-05 20:49  Reset password (by admin)                  success  by sabithaskaranavar@<tenant>
    2026-11-05 20:51  Update PasswordProfile                     success  by Microsoft password reset service
    2026-11-05 20:51  Update user                                success  by Microsoft password reset service
    2026-11-05 20:51  Update PasswordProfile                     success  by Microsoft password reset service
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    2026-11-05 20:52  Update user                                success  by Azure Credential Configuration Endpoint Service
    2026-11-05 20:52  User registered security info              success  by sarah.chen@<tenant>
    2026-11-05 20:52  Update user                                success  by Azure Credential Configuration Endpoint Service

  PASS  identity verified before the reset
  PASS  old MFA methods removed at reset
  PASS  new MFA method registered after the reset
  PASS  audit: admin deleted security info
  PASS  audit: user registered security info
  PASS  account still enabled

6/6 checks passed
```

Figure 6: Verify: new method registered after the reset, audit trail, all checks passed

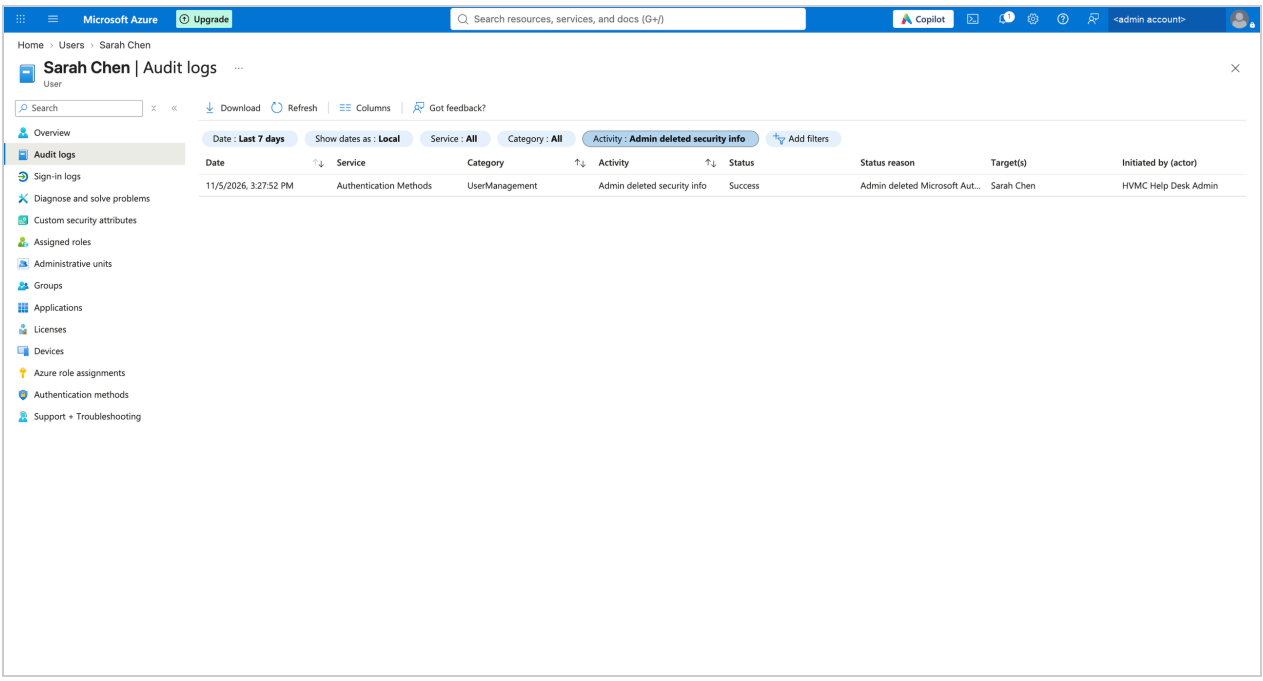


Figure 7: Entra admin center audit log: “Admin deleted security info” (Microsoft Authenticator) at 3:27 PM, initiated by HVMC Help Desk Admin, not the Global Admin

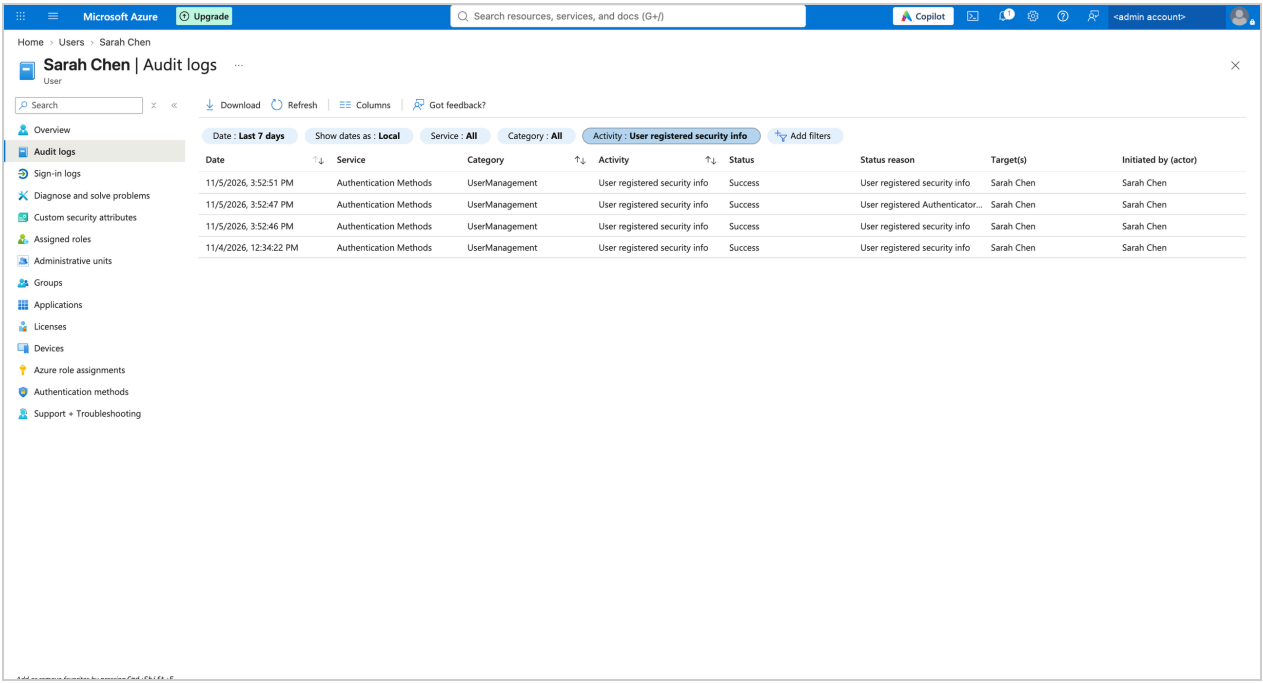


Figure 8: Entra admin center audit log: “User registered security info” by Sarah Chen: the old phone on 4 Nov and the new phone on 5 Nov at 3:52 PM

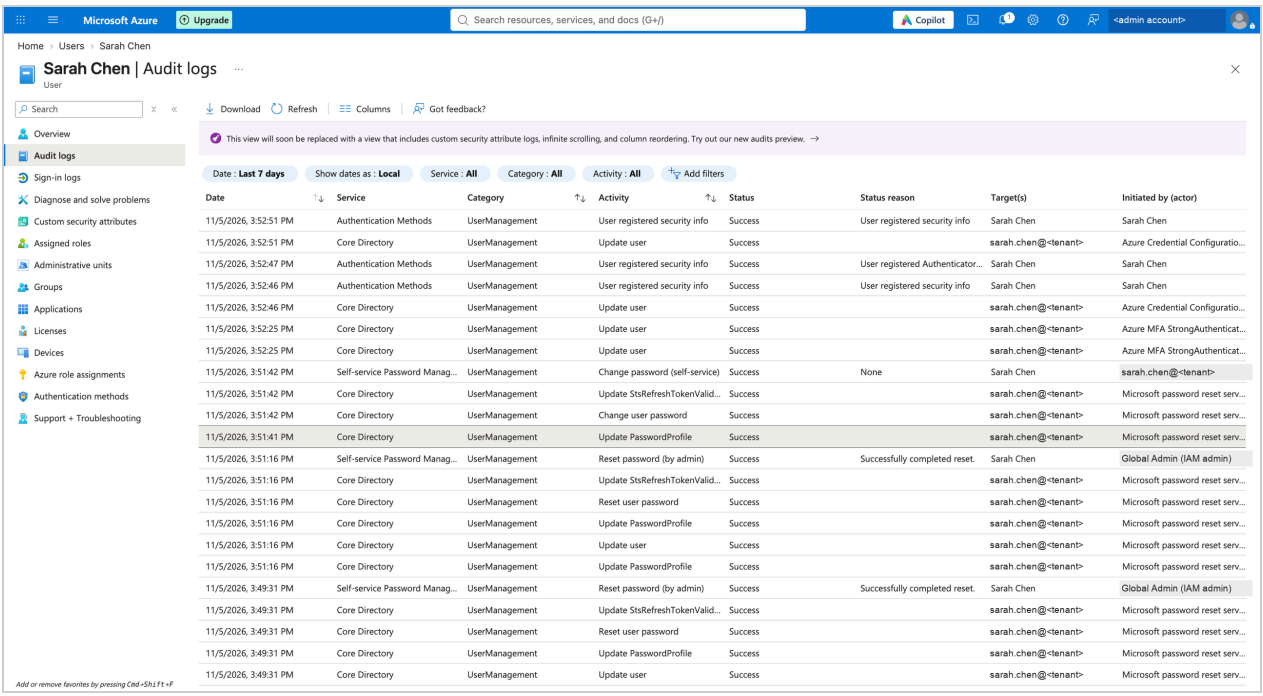


Figure 9: Full audit trail for Sarah Chen on 5 Nov (domain and admin e-mail masked)